

# Bounded Communication Fault Tolerance

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# Introduction

## Today:

- ▶ Bounded Communication Fault Tolerance
  1. Walkthrough the math
  2. Alternative definition to Expander decomposition
  3.  $(d, k)$  - Toric
- ▶ New experimental results.
- ▶ Answering the code teleportation.

# Bounded Communication Fault Tolerance

## Setting

We allow a non-nosy, strong as we wish, classical computation aside. Yet we would like to limit the communication to a constant number of bits per logical qubit, in each round.

## Justification

1. If we don't have a constant overhead in depth, in that model,  
 $\Rightarrow$  we also don't have it in the standard model.

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2. In that model, gate teleportation for a constant dimension code is easy. Namely, for the Topological codes the task is reduced to finding a decoding with a constant-size hint.

# Max Color

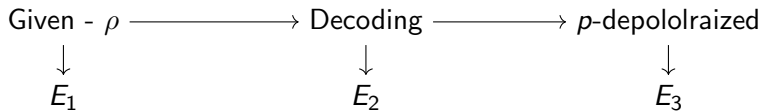


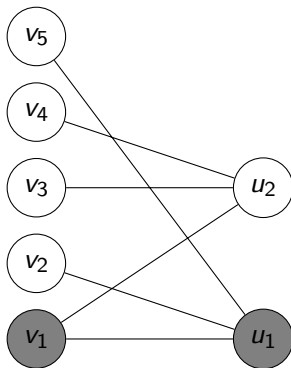
Figure: Illustration of the cycle.

standard model,  $t = 0$

$u_2$

$u_1$

simplified  $(p', q)$ -model



## What did we learn?

1. Storing time is, likely, not  $\Theta(e^n)$ , might be  $\Theta(L^\alpha)$  for  $\alpha \in (0, 1)$ .
2.  $\Rightarrow$  Implies that the structure of the theorem/key-lemma is different than the known refrigerators.

# Future

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  - ▶ More experiments, higher statistical standards?
  - ▶ Accelerate the decoder? Shifting to *c/c++*? Parallelize it with GPUs? Do we have budget?



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3. How to use the fact our code is a quotient code?